

Abstract

Among all the challenges that face Egypt, urban congestion, pollution, public health issues, and population growth are being focused on this semester. These problems are connected, as each challenge enhances the other. These are the reasons for the main problem, which is Traffic congestion and its consequences in highly concentrated squares in Egypt. This problem led to an increase in the accident rates and a rise in pollution. Searching for previous solutions for inspiration, LOS ANGELES -ATSAC (automated traffic surveillance and control) and Global Position System (GPS) have worked on solving this problem. While considering their positives and negatives as projects. These are the reasons for constructing the innovative solution to organize the intersections for less wasted time for people time. The solution is a smart traffic system that utilizes V2V communication to achieve a lower response time, thereby helping to reduce traffic flow waves. The system provides a smart traffic light in the intersection modified with a camera connected to an AI model that decides which lane to go first in a specific time, which is better than the normal system that takes the same time for all the lanes empty or full. In addition, the green corridor that opens with a fence by a hydraulic lift enhances the safety procedures and provides emergency services. The green corridor works for the ambulance and emergencies only, and it works as an extra lane during traffic hours. By following some design requirements for constructing the solution. Some parameters were chosen to be measured for data collection. The results are illustrated as 94% system detection accuracy on average, a total response time of 9.313 seconds for each cycle, and an increase in the system's efficiency of 28%. These results show the efficiency of the idea and prove that the solution that is presented is the best solution for traffic congestion.

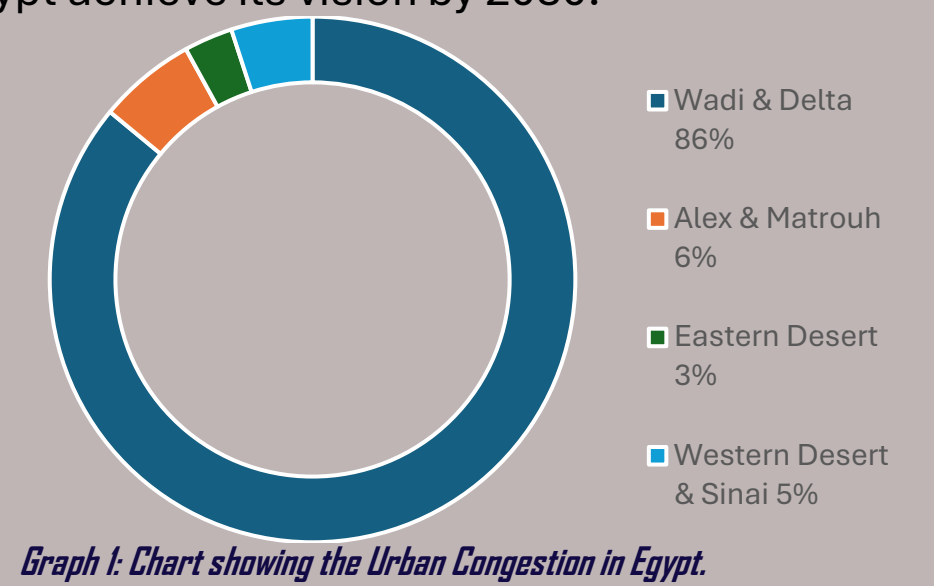
Introduction

Egypt has been suffering from its grand challenges since the last century; those grand challenges impede causing obstacles for Egypt while rising to the top. The solution is to solve one of those problems to help Egypt achieve its vision by 2030.

One of the challenges Egypt is facing is the traffic congestion problem. The road in any country is like its arteries, veins, and the vitality of cities depends on it primarily. By solving this problem, there would be fewer accidents and less chaos on the roads. People would save much more time and fuel every day, and the air would be cleaner because cars wouldn't stay stuck for hours.

Egypt would look more modern, and people would enjoy moving around the city. The main problem consists of these challenges, and to solve it, some design requirements were stated to be met. Decreasing the time waiting at intersections, decreasing the percentage of accidents, and lowering the response time for the system to take action.

Two prior solutions were considered while searching for the solution; the first is LOS ANGELES–ATSAC (automated traffic surveillance and control). One of the largest and most advanced smart traffic



Graph 1: Chart showing the Urban Congestion in Egypt.



Figure 1: ATSAC (prior solution) control panel.

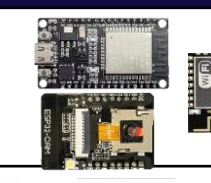
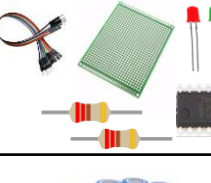
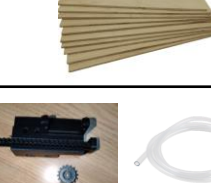
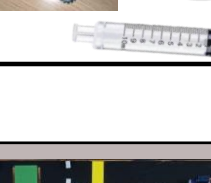
The second prior solution is the It is a satellite-based navigation system invented in the 1970s, and it works through a network of about 30 satellites that orbit the Earth and send signals continuously to receivers as phones. GPS is used to know the exact location of a person or a vehicle, and it is efficiently used in decreasing urban congestion problems in many ways.

After extensive research and analysis of prior solutions and our subject's learning outcomes, we deduced the solution that consists of three parts. The first is that the traffic signs will be timed according to a ration between the density of cars in each road in the intersection; cameras, ultrasonic sensors and AI system will be used to operate the system; counting the cars and using the time of varying the traffic light color will be calculated according to the cars count, this will help in rush hours, as there will be more dense road than the other roads in the intersection; by $(G_t = (C_t - L_t) * \text{road cars} / \text{total intersection cars})$, where G_t is the green time for the road, C_t is the total cycle time, and L_t is the total lost time or yellow time in the intersection. According to this, more dense roads will have a longer time in the green light. Constant distance between each car.

The second part is maintaining a constant distance between each car so that if any car stops, there will be enough distance for the following cars to take action. Also, using V2V communication technology, the following vehicle will be alerted if the front car presses the brake. Thirdly, is the Emergency Road (green corridor). It will be used as an extra lane for emergencies, such as for ambulances and police cars. Additionally, during rush hour and if there is maintenance work on any lane, it will be utilized. In the normal traffic state, this lane will be closed using a roadblock to prevent unauthorized use.

Materials & Methods

Table 1: Table of materials.

Item	Description	Usage	Source	Quantity	Total Cost	Picture
Micro controllers	Micro controller and a camera and controllers with Wi-Fi abilities	Controlling all the system, and car detection system, and V2V communication	Electronics Shop	ESP32 *1 ESP32-CAM *1 ESP8866 *3	1330 LE	
Electric components	Electronic components that controls the traffic system by sending signals and illustrate it	Controlling the LEDs, representing the traffic lights, adjusting the current in the system, and connecting all them	Electronics Shop	220Ω resistors *12 LEDs *12 PCB breadboard *1 Shift register IC *2 Jumper wires *50	94 LE	
Batteries	Rechargeable 2000 milli-amps lithium-ion batteries	Provides the whole system with power	Electronics Shop	Batteries *7	350 LE	
Servos, stepper, and DC motors	Motors that rotate by the signals of microcontrollers	Moving different parts in the prototype	Electronics Shop	Stepper *1 Sg 90 micro servo motor *3 DC geared motor *3	375 LE	
Wooden sheets	100*120 cm - 3mm	Works as a base for the prototype construction	Carpenter	Wooden sheet *1	160 LE	
Syringe + soft tube + linear actuator	10 cm syringe + 5 mm plastic tube + 3D printed gear system	Parts that control the hydraulic lift for the green corridor fence	Pharmacy + 3D printed	Syringes *6 Soft tube *1 Linear actuator *3	200 LE	
TOTAL COST					2508 LE	

Methods of construction:

-A black sticker was glued to the wooden sheet, and a white sticker was used to give the shape of the road.
A laser cutter is used to make the fence's shape, cutting the holes for the wires, making the traffic light poles, and making a box for the lithium battery of the ESP camera
PCB breadboard connected to a DC motor, ESP8266, ultrasonic sensor, and battery to make the vehicle.
-All the electronic components were soldered on a PCB bread board including (esp32, servos, stepper, shift registers, resistors, power source) to make the prototype's motherboard.
-The code for the esp32 was uploaded to control all the components through Wi-Fi server HTTP requests.
-An ESP32 cam was installed at the middle of the road intersection above a 2d cut box carrying 2 lithium batteries for power providing, attached to the stepper motor.
-A model that is trained to detect the colors of the toy cars from images was built.
Using OpenCV library in python.
-A simple Python code was built to take the output from the AI car detection model and calculate the green time according to a specific equation.
-A Python desktop application was built to control the motherboard using the added endpoints according to the output of the integration between the car detection model and the green time calculation model.
-The last step is installing the motherboard and all its components into the prototype parts.

Test plan:

The prototype was tested to achieve four design requirements, which are AI detection accuracy for cars, the total response time for the system and the AI run-time, and the system's efficiency compared to a normal system in terms of traffic intersection flow rate. However, after testing the prototype following the requirements, the results are finally shown.

Results

The prototype was mainly tested in terms of the following 4 design requirements:

-Image car detection model accuracy -Traffic system HTTP requests response time
-Efficiency of increase in intersection flow rate -V2V communication system efficiency

1) Car detection accuracy:

The model was tested on detecting cars in the 4 intersection roads in 4 intersection cycles. The total car count in the image was 50, and the model was able to detect 47 of them with a detection accuracy of 94%.
2) Traffic system HTTP requests response time:

Each request was tested on 4 trials, and the average was calculated from them.

Request	Request Type	Data transferred	AVG Response time
ESP32 cam image capturing	GET	Image	264 ms
Stepper motor rotating order	POST	JSON raw input	1.15 sec
Servo motor angle control order	POST	JSON raw input	177.8 ms
LEDs status updates order	POST	JSON raw input	255.3 ms

3) Efficiency of increase in intersection flow rate:

Trial I	(Rf)	(N)	current systems Green Time	Ratio based Green time	Rt * current systems G _t	Rt * ratio based G _t	Min (Road Rt * current systems G _t , N)	Min (Road Rt * Ratio based G _t , N)
Table 4: Green light time for normal and AI-based systems.								
Road 1	0.333	5	10 sec	18.2 sec	3.33	6.06	5	3.33
Road 2	0.333	4	10 sec	14.5 sec	3.33	4.82	4	3.33
Road 3	0.333	1	10 sec	3.6 sec	3.33	1.19	1	1
Road 4	0.333	1	10 sec	3.6 sec	3.33	1.19	1	1
Σ							11	8.6
System's Efficiency								
(11/8.6) * 100% = 128%								

This illustrates that the system is able to increase the number of cars that can pass through the intersection by 128%

Analysis

The solution "Solving Traffic Congestion Using AI-based traffic lights management and V2V Communication" works to solve two main challenges, which are urban congestion and dealing with population growth. These challenges are causing the main problem, which is traffic congestion in Greater Cairo intersections. As the Greater Cairo Region (GCR) is known to be represent about 19% of Egypt's population, as it had about 16.1 million inhabitants in 2018, and this number has grown in the past few years.

After long research, a solution has been reached, which is an AI-based traffic system that works with three main programs. A traffic light that works with the ESP32-CAM, with an AI model to count the number of cars to detect the priority of the busiest road, and give the exact time for each road to be open to save people's time. This is decided by the AI model, which works by inspiration from CH.3.01 The AI model works in the same steps of the scientific method. As it learns to work, following the observed patterns in data and develops predictive "laws." For instance, it might deduce that traffic congestion increases. In addition to a green corridor for emergencies that works with a hydraulic lift system, just like in B1.3.05 reflexes in muscles are responsible for any emergency that needs a speedy response and takes action. The same thing happens in the prototype; if any danger occurs that affects traffic, the green corridor opens when a lane is busy, or there is an accident, or even for an ambulance to move faster alone in that extra lane. Not forgetting to mention the V2V communication to keep a safe distance between the cars and avoiding eh late response time of humans.

After analyzing how the idea works, some design requirements were tested to determine whether they were achieved or not. The traffic light system was tested to determine the time that the intersection takes in one cycle. Making four trials, changing the number of vehicles each time to check the system detection accuracy, the result was that the average was calculated, and the average detection accuracy equals 94.2% this was estimated by the law:

$$\text{Average detection accuracy} = (\text{total detected cars} / \text{total tested cars}) * 100$$

After getting the system accuracy as shown in Graph 2, the first trial got 100% accuracy, the others had the problem of that the AI model couldn't detect some colours like black, so it miscounted them considered as a negative result. However, by analyzing the first trial, which was the most accurate. The intersection flow rate was calculated by getting the time that each car passes the intersection, and comparing the normal system green light time to the AI-based system, as shown in Graph 3. The difference in allocation of

time for all four roads in the intersection was based on the number of cars, resulting in less time wasted by people. This comparison led to the system efficiency, which was calculated by the traffic flow rate, according to this equation:

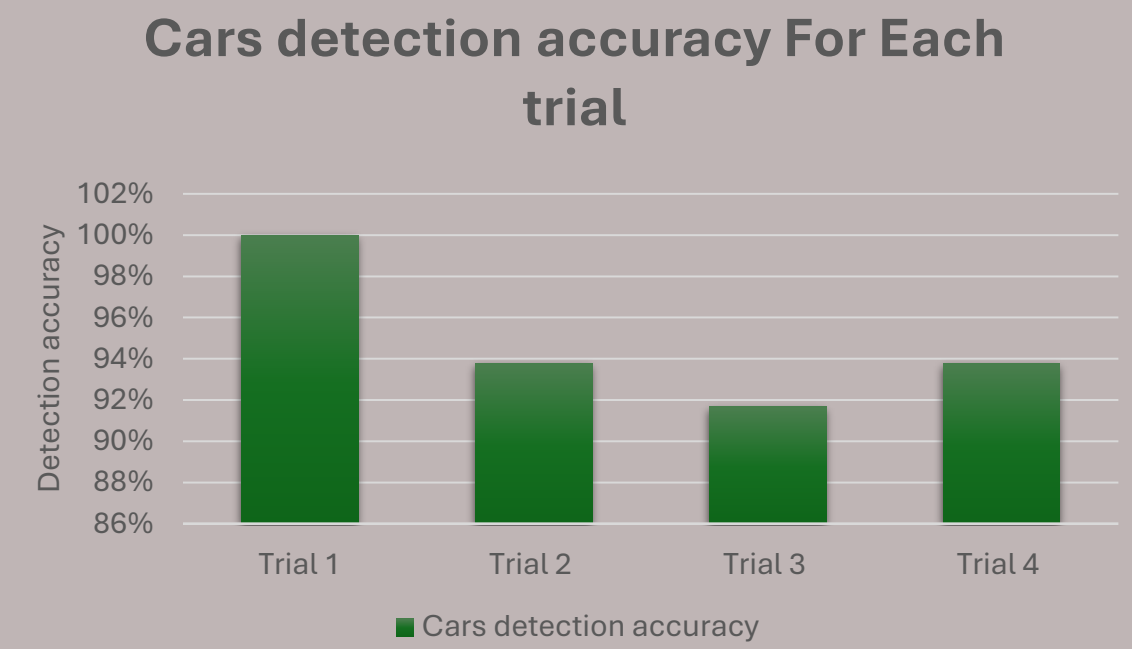
$$\sum_{r=1}^4 \min(\text{Road flow rate} * \text{ratio based Green Time}, \text{road cars count})$$
$$\sum_{r=1}^4 \min(\text{Road flow rate} * \text{current systems Green Time}, \text{road cars count})$$

With that, the system's efficiency is 128% which is higher than the normal system with 28% in this case, as the traffic flow rate in this condition is 0.333 for each road. These results illustrate the time that is saved by the system as each road gets the exact time to be free of cars. As the first road had 5 cars, it needed 18.2 seconds to pass all the cars through the intersection, but in the normal state, the green light would take 10 seconds for each road, so the difference is shown in how to organize the priority and calculate the Green Light Time. This equation was used:

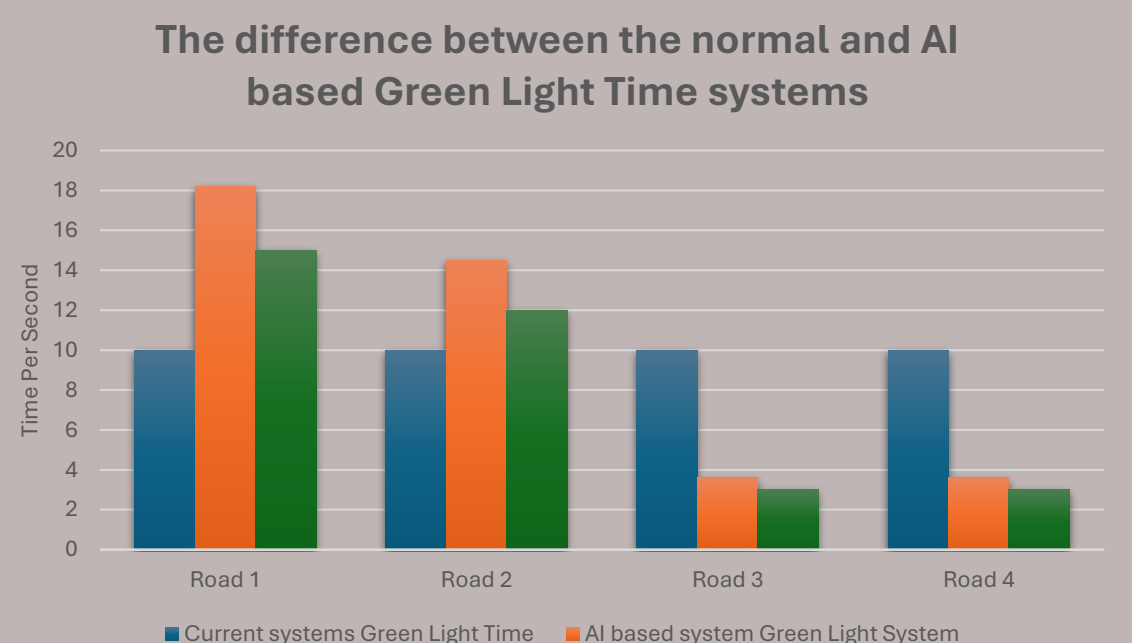
$$G_t = (C_t - L_t) * \text{road cars} / \text{total intersection cars}$$

Where C_t is the total Cycle time, and L_t is the lost time or the yellow time.

To continue, the third parameter calculated was the system time response. The time of the system working for each cycle was divided into two parts: the response time of the devices and the AI model runtime to receive data, analyze it, and send orders. As studied in PH.3.04 The IoT system mirrors this exactly: sensors (transmitters), wireless or fiber channels (medium), and central AI servers (receivers). This structure ensures real-time, error-free communication between devices. In addition to using antennas that allow V2V communication, enhancing the data range. For the calculations of the system response time, a desktop Windows application called Postman was used to test various APIs and endpoints of the system, providing accurate response time for each request. To calculate the total time that the cycle needs to empty four roads with no errors. The detected response time of image capturing, stepper motor rotating, the LEDs status updating, and the green time calculation in the previous part. Adding the run time of the AI model to receive data from the ESP32-CAM and the counting process, and the order sending to the lights. The total time response is calculated to be 9.313 seconds.



Graph 2: Cars detection accuracy for each trial.



Graph 3: The difference between the normal and AI-based green light time systems.

Conclusion

The idea showed how it functions and how it can help in solving Egyptian grand challenges and the big problem of traffic congestion by decreasing the time waiting in intersections and the random stopping of the cars, showing great results as follows:

- The system is able to increase the number of cars that can pass through the intersection by 128%
- An efficient system to avoid emergency cars from road congestion by creating a green corridor.
- High efficiency in decreasing the percentage of accidents by applying V2V communication.
- Controlling the congestion and solving it in the least time, as the response time for the system in each cycle is only 9.313 seconds to empty the roads.
- Increasing the activity and the vitality of Egypt's roads.

The project shows that it can get higher results as a life application than the prior solutions, like the ATSAC, not just achieving the design requirement but also exceeding it, and providing a perfect solution for the main problems Egypt's roads face.

Recommendations

"Nothing is perfect by humans." By the time new developments and technologies emerge, they often affect existing ideas or projects. This is why it's essential in scientific research to write recommendations for others or researchers who will work in the same field or on the same project, thereby avoiding mistakes.

- Using a wide lens camera, such as a traffic enforcement camera, for the road so more vehicles can be detected, and the intersection can move faster.
- Applying the V2V for more applications, like: keeping a safe distance, sending direction signals early, and issuing warnings about sudden breaks.
- Connecting the system to signal towers along the road so all the cars get the updates faster instead of the router that works as the receiver and server at the same time.
- Using the Yolo 8 AI model for a better real car detection accuracy, leaving no errors.
- Enhancing the security of the system by specifying access to specific devices' credentials.
- Using a smaller sample that looks like a real car for a better simulation process, specifying the kinds of vehicles like microbuses and tok-toks, so the AI model can detect the different types and count them.

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